

EXP.NO: 5 EDA – DATA VISUALIZATION

AIM:

To explore and understand the underlying patterns, distributions, and relationships within the dataset through visual representations, which aids in uncovering insights, detecting anomalies, and guiding further analysis.

PROGRAM

```
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
```

```
data = {
    'Category': ['A', 'B', 'C', 'D'],
    'Values': [3, 7, 5, 4]
}
```

```
df = pd.DataFrame(data)
print(df)
```

```
x = np.linspace(0, 10, 100)
y = np.sin(x)
```

```
plt.plot(x, y, label='Sine Wave')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.title('Line Plot')
plt.legend()
plt.show()
```

```
categories = ['A', 'B', 'C', 'D']
values = [3, 7, 5, 4]
```

```
plt.bar(categories, values)
plt.xlabel('Category')
plt.ylabel('Values')
plt.title('Bar Plot')
```

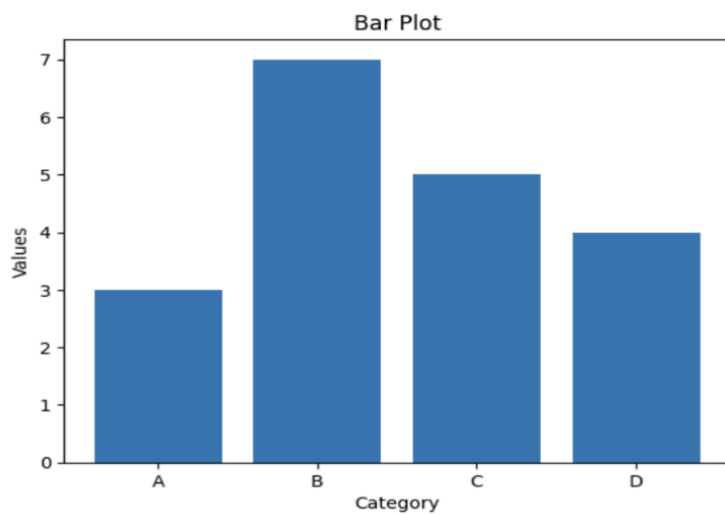
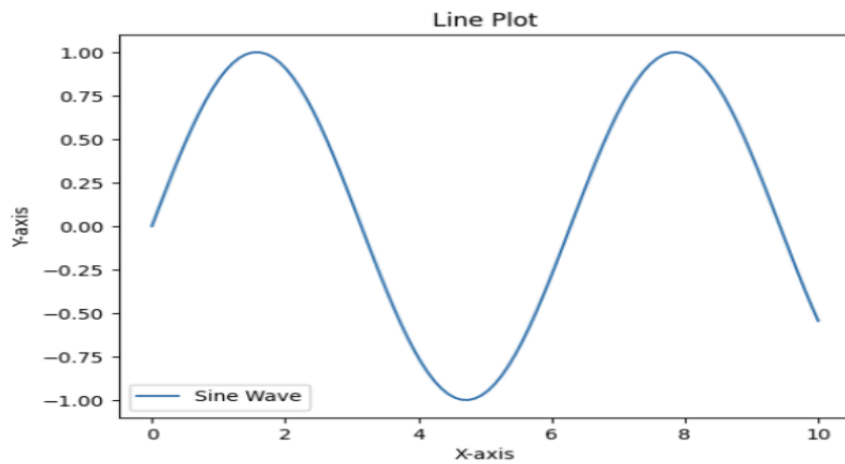
```
plt.show()
```

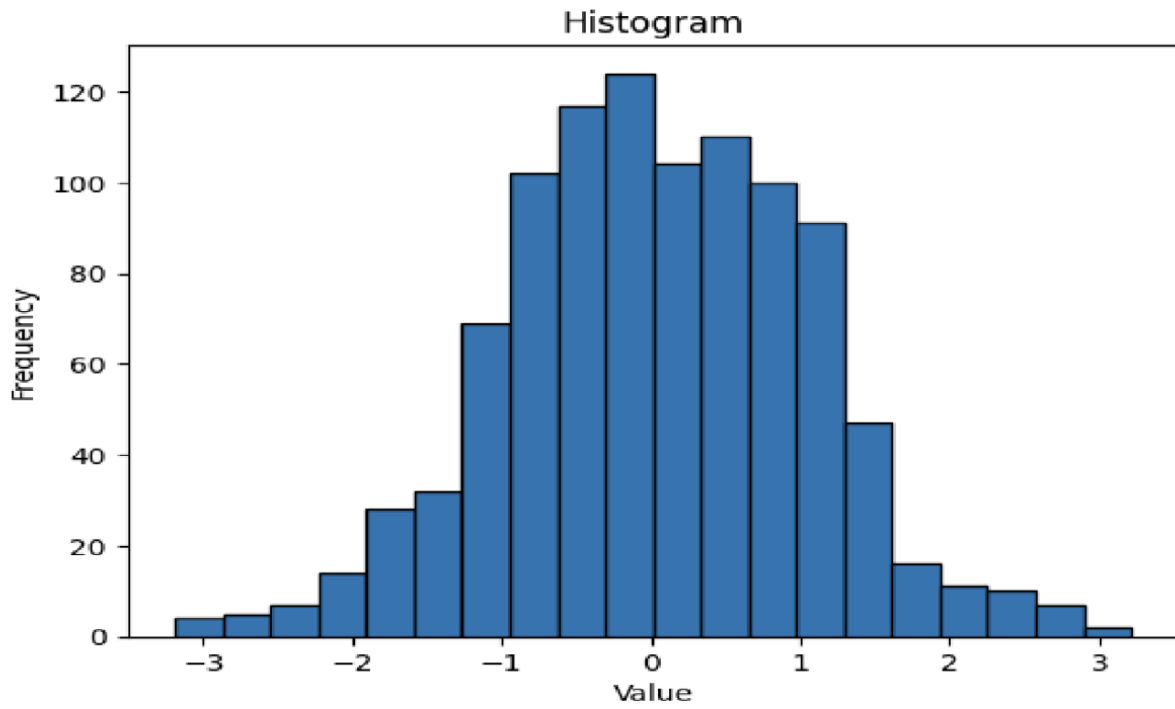
```
data = np.random.randn(1000)
```

```
plt.hist(data, bins=20, edgecolor='black')  
plt.xlabel('Value')  
plt.ylabel('Frequency')  
plt.title('Histogram')  
plt.show()
```

OUTPUT:

	Category	Values
0	A	3
1	B	7
2	C	5
3	D	4





RESULT:

Thus, the given program was written and executed successfully.